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MA3521 — $\mathbf{y}' = A\mathbf{y} + \mathbf{f}(t)$ Exponential and Trig (II)

Name:

1. Solve $\mathbf{y}' = A\mathbf{y} + e^{-2t}\begin{pmatrix}-7\\-7\end{pmatrix}$, $\mathbf{y}(0) = \begin{pmatrix}4\\0\end{pmatrix}$ and $A = \begin{pmatrix}2 & 3\\3 & 2\end{pmatrix}$ with epairs 5, $\begin{pmatrix}1\\1\end{pmatrix}$; -1, $\begin{pmatrix}1\\-1\end{pmatrix}$

   (a) $\mathbf{y}_c(t) =$

   (b) $\mathbf{y}_p(t) =$ is a suitable UC form.

   (c) Row reducing gives the UCs in

   $\mathbf{y}_p(t) =$

   (d) Row reducing gives constants $\mathbf{c}$ in

   $\mathbf{y}(t) =$

2. Solve $\mathbf{y}' = A\mathbf{y} + \cos(2t)\begin{pmatrix}5\\5\end{pmatrix}$, $\mathbf{y}(0) = \begin{pmatrix}0\\-2\end{pmatrix}$ and $A = \begin{pmatrix}2 & -1\\3 & -2\end{pmatrix}$ with epairs $1, \begin{pmatrix}1\\1\end{pmatrix}$; $-1, \begin{pmatrix}1\\3\end{pmatrix}$

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(b) $\mathbf{y}_p(t) =$ is a suitable UC form.

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